MS&E 226 — Midterm 1 Practice Questions (Generated Set 1): Solutions

Question 1 — Answer: (a)

Average MSE = irreducible error + Bias^2 + Variance $= 9 + (-2)^2 + 5 = 9 + 4 + 5 = 18$.

(a) Correct — the arithmetic above.

(b) Uses Bias instead of Bias^2: $9 + (-2) + 5 = 12$. The decomposition requires squaring the bias; a negative bias still costs you positive MSE.

(c) Drops the irreducible error term entirely ($(-2)^2 + 5 = 9$) — this is the **estimator** MSE formula from Lecture 8 ($\text{MSE} = \text{Bias}^2 + \text{Variance}$, no irreducible error), not the **prediction** MSE formula from Lecture 7. Mixing up the two decompositions is the single easiest way to get this kind of question wrong.

(d) Confuses MSE with RMSE by taking a square root that doesn't belong here.

Question 2 — Answer: (c)

(a) Misreads "decomposition" as "fixed total." Nothing constrains Average MSE to stay constant — a learning algorithm can simply get better (lower MSE) by improving bias and variance simultaneously.

(b) The Gauss-Markov theorem is real, but it only applies to a specific comparison: linear, **unbiased** estimators using a **fixed** feature set. It says nothing about comparisons across different feature sets or non-linear algorithms, which is exactly the loophole double descent and the weight-measurement example exploit.

(c) Correct. The bias-variance decomposition, Average MSE = irreducible error + Bias^2 + Variance, is exactly that — a decomposition of whatever MSE happens to be, not a conservation law that forces the pieces to trade off. Lecture gave concrete counterexamples: the repeated weight-measurement example (both bias and variance fell as more measurements were added) and the double-descent lasso example (adding genuinely informative features reduced both bias and variance, and even reduced the "irreducible" error relative to the enlarged feature set).

(d) Overcorrects to the opposite blanket claim. Bias and variance don't always move together either (e.g., regularization's whole point is moving them in opposite directions) — there's simply no universal law either way.

Question 3 — Answer: (b)

(a) Contradicts the Gauss-Markov theorem directly: no unbiased linear estimator can beat OLS's variance.

(b) Correct. Under (A1)–(A3), the Gauss-Markov theorem says OLS has the **lowest variance among all linear, unbiased estimators** that use the same features. If the ridge estimator (which is linear in $\boldsymbol{Y}$) has

strictly lower variance than OLS, it cannot be unbiased — otherwise it would contradict OLS's optimality within that class. So the ridge model must have nonzero bias somewhere.

(c) Ignores Gauss-Markov. This is the right instinct when comparing across **different feature sets** or **different algorithm families** (Question 2′s lesson), but Gauss-Markov gives a precise, forced relationship for **this specific** comparison (ridge vs. OLS, same features, under A1–A3) — context matters.

(d) Backwards. Regularization's mechanism is to **add** bias in exchange for variance reduction, not to "correct" an existing OLS bias (OLS already has zero bias under A1–A3, so there's no bias to correct).

Question 4 — Answer: (c)

(a) Gauss-Markov's zero-bias guarantee requires (A2) — **all** relevant features observed. Omitting C violates this, so the guarantee simply doesn't apply here.

(b) Directly contradicts the lecture's point that the bias "depends on how correlated the remaining features are with the omitted features."

(c) Correct, per the lecture's remarks on omitted-variable bias (OVB): omitting a relevant feature breaks assumption (A2) (not all relevant features observed), so the Gauss-Markov zero-bias guarantee no longer applies, and both the size of the resulting bias and the distortion to the fitted coefficients on the **included** features depend on the correlation between the omitted and included features.

(d) Contradicts the omitted-variable-bias (OVB) phenomenon explicitly named in lecture: omitting a correlated feature can corrupt the **coefficients** on the included features too, not just the final predictions.

Question 5 — Answer: (c)

(a) False in general — plenty of estimators are biased (e.g., OLS with an omitted feature, from Question 4). Being unbiased isn't why the irreducible-error term disappears; the reason is about what kind of thing the **target** is (random outcome vs. fixed parameter), not about the estimator's own bias.

(b) Invents a restriction that isn't in the material — irreducible error is a general prediction-theory concept, not classification-specific.

(c) Correct. Irreducible error in the prediction setting exists because the **target** Y is itself a random variable with leftover variance even given (x) exactly (i.e., $\sigma^2((x)) = \mathrm{Var}(Y \mid (x)) > 0$ in general). In the estimation setting, the target is a population parameter θ, which the frequentist view treats as a **fixed**, non-random quantity — there's no analogous leftover randomness in a fixed number, so there's nothing for an "irreducible error" term to capture.

(d) Confuses two different "shrinks with n" facts: it's the **variance** of an estimator that shrinks as n grows, not irreducible error (which, in the prediction setting, is about the conditional variance of Y given (x) and has nothing to do with how much data you have).

Question 6 — Answer: (c)

(a) Reverses the frequentist view (closer to how some people informally imagine a Bayesian perspective, but even that isn't quite this).

(b) Treats the parameter as random too, which the frequentist view explicitly rejects.

(c) Correct — this is the core frequentist premise: the true parameter θ is fixed and unknown; the only randomness is in which particular data sample you happened to draw (the "parallel universes" thought experiment), and any estimator computed from that sample inherits its randomness.

(d) A common but mistaken intuition: once you're holding a specific, already-collected dataset, the **numbers in front of you** are fixed, but that doesn't mean there was no randomness in the process that produced them. The parallel-universes framework reasons about the data-generating process, not about a single realized dataset in isolation.

Question 7 — Answer: (a)

$\mathrm{SE} = \frac{\hat{\sigma}}{\sqrt{n}}$, so the ratio of standard errors is the **inverse square root** of the ratio of sample sizes:

$$\frac{\widehat{\mathrm{SE}}_{25}}{\widehat{\mathrm{SE}}_{400}} = \frac{\sqrt{400}}{\sqrt{25}} = \frac{20}{5} = 4.$$

So $\widehat{\mathrm{SE}}_{25}$ is about 4 times larger than $\widehat{\mathrm{SE}}_{400}$.

(a) Correct, by the computation above.

(b) Uses the ratio of sample sizes directly ($\frac{400}{25} = 16$) instead of the ratio of their square roots — forgetting that SE scales with $\frac{1}{\sqrt{n}}$, not $\frac{1}{n}$.

(c) Confuses the standard deviation of individual observations (which doesn't shrink with n) with the standard error of the sample **mean** (which does, proportionally to $\frac{1}{\sqrt{n}}$) — precisely the distinction lecture called out as important to keep straight.

(d) Gets the direction backwards: **more** data means a **smaller** standard error, not a larger one, so it's $\widehat{\mathrm{SE}}_{25}$ that's bigger.

Question 8 — Answer: (b)

(a) This is exactly the tempting-but-incorrect interpretation flagged in lecture: it treats μ as if it were still random after the interval has been computed.

(b) Correct. A confidence interval's coverage is a statement about the **procedure**, evaluated across repeated samples (parallel universes): 95% of the intervals this procedure produces will contain the true μ. Once you've plugged in one actual sample and computed one specific interval, there's no more randomness left in μ to assign a probability to — μ is fixed; the interval (built from your random sample) was the random object, before you computed it.

(c) Misapplies the 95% figure to $\overline{Y}$ instead of μ — and doesn't even make sense on its own terms, since a symmetric CI is centered exactly on the observed $\overline{Y}$, which trivially lies at the interval's midpoint, not in it "with 95% probability."

(d) Not required: the Central Limit Theorem guarantees the sampling distribution of $\overline{Y}$ is approximately normal for large n **regardless of the shape of the individual Y_i's distribution** — that was explicitly called out in lecture as the "remarkable" part of the CLT.

Question 9 — Answer: (a)

(i) is correct. A higher confidence level requires a larger critical value ($z_{\alpha/2} \approx 2.58$ for 99% vs. 1.96 for 95%), which directly widens the interval $\overline{Y} \pm z_{\alpha/2}\widehat{\text{SE}}$, holding n and $\hat{\sigma}$ (and hence $\widehat{\text{SE}}$) fixed.

(ii) is incorrect. "Guaranteed to contain μ" describes neither interval. Both are built from the same random sample and both have only a long-run **coverage rate** (99% or 95%, respectively, across repeated sampling) — not a guarantee for this one specific interval. This is the same misconception as Question 8, applied to a comparison between two intervals instead of one.

Since (i) is correct and (ii) is not, the answer is (a).

(a) Correct — see the reasoning above.

(b) (ii) alone is not correct; see above.

(c) (ii) is not correct, so both cannot be.

(d) (i) is correct, so it isn't true that neither is.

Question 10 — Answer: (b)

$\overline{Y}$ is literally the same number, computed the same way, in both analyses — so its Bias and Variance (as derived in Lecture 7 for the sample mean specifically) are identical regardless of which question you're using it to answer: $\text{Bias} = 0$ and $\text{Variance} = \sigma^2/n$. What differs is only the **target** being compared against:

- As a **prediction** of a new outcome Y (Lecture 7): Y itself is random even once you know μ exactly, contributing an irreducible error term σ^2. Average $\text{MSE} = \sigma^2 + 0 + \sigma^2/n$.
- As an **estimator** of μ (Lecture 8): μ is a fixed parameter under the frequentist view, so there's no analogous leftover randomness to contribute an irreducible term. $\text{MSE} = 0 + \sigma^2/n$.

(a) False — $\overline{Y}$ doesn't change formula between the two uses; it's the exact same statistic playing two different roles.

(b) Correct, matching the reasoning above.

(c) Ignores the irreducible error term entirely; the two MSEs differ by exactly σ^2 (see Question 5's reasoning for why that term appears in one case and not the other).

(d) Garbles which term is "the same" (it's Variance **and** Bias that match, not just Variance) and misattributes irreducible error to Bias — irreducible error is a separate, third term in the decomposition, not a form of bias.